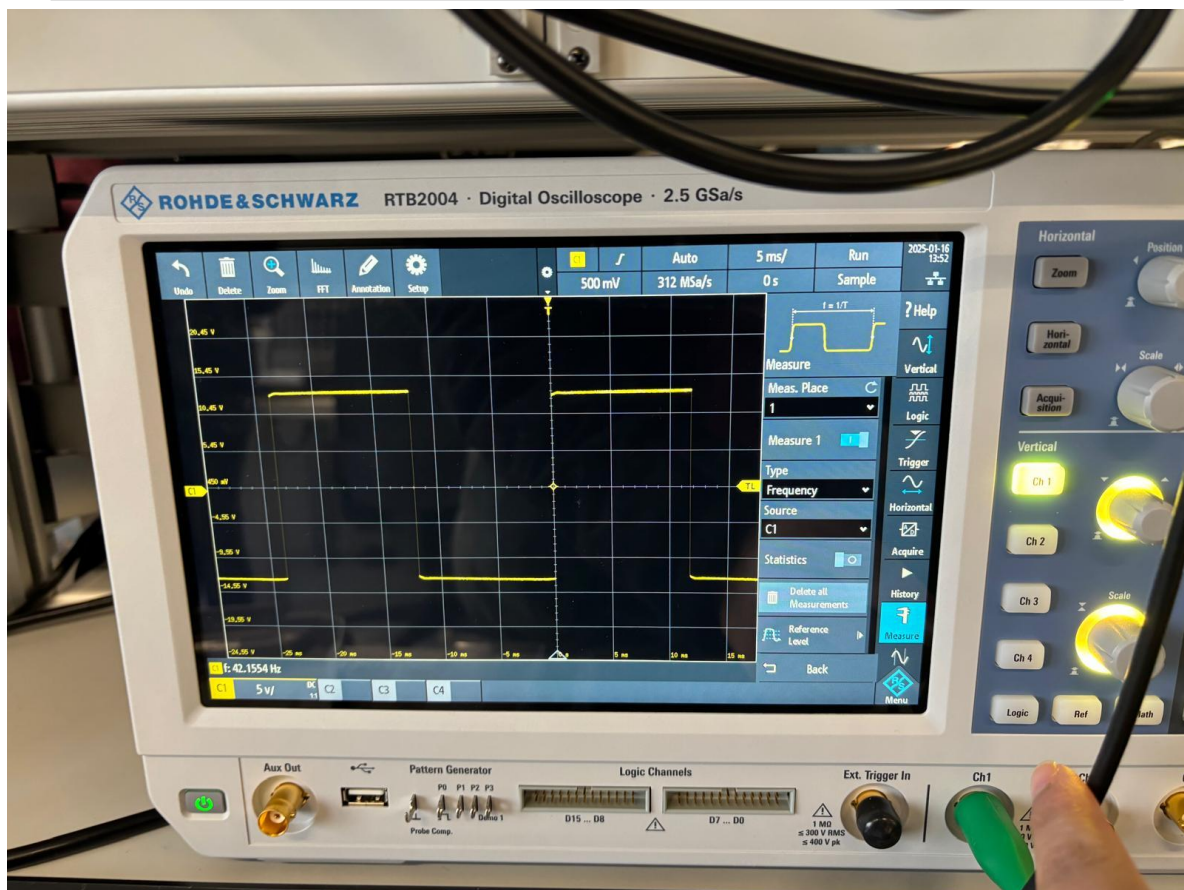


Measurement Parameters Organized by Task

Task	Parameter	Description	Measured Value
1	f	Expected frequency of the square wave.	42.1554 Hz
1	V_{pp} of V_{out}	Peak-to-peak voltage of the output signal measured when $f = 25$ Hz (square wave).	26.7 V
2	V_{pp} of V_{out}	Peak-to-peak voltage of the output signal measured when $f = 50$ Hz (triangular wave).	17.4 V
3	V_{pp} of V_{out}	Peak-to-peak voltage of the output signal measured when $f = 50$ Hz (PWM).	27.5 V
3	Duty Cycle (%)	Duty cycle of the PWM signal.	27.5%



Task 1: Square Wave Generation

Observations:

- Oscilloscope Display (Image 1):
 - Output shows a clean square waveform with distinct high and low voltage levels.
 - Frequency: Approximately $f \approx 42.1554 \text{ Hz}$.
 - Voltage Levels: $V_{\text{high}} = +5 \text{ V}$, $V_{\text{low}} = -5 \text{ V}$.

Analysis:

- The frequency was verified using the oscilloscope's measurement tools.
- The resistor-capacitor time constant directly determines the square wave's frequency:

$$f = \frac{1}{2R_{\text{eq}}C}$$

- Any deviation from the theoretical value can be attributed to component tolerances.

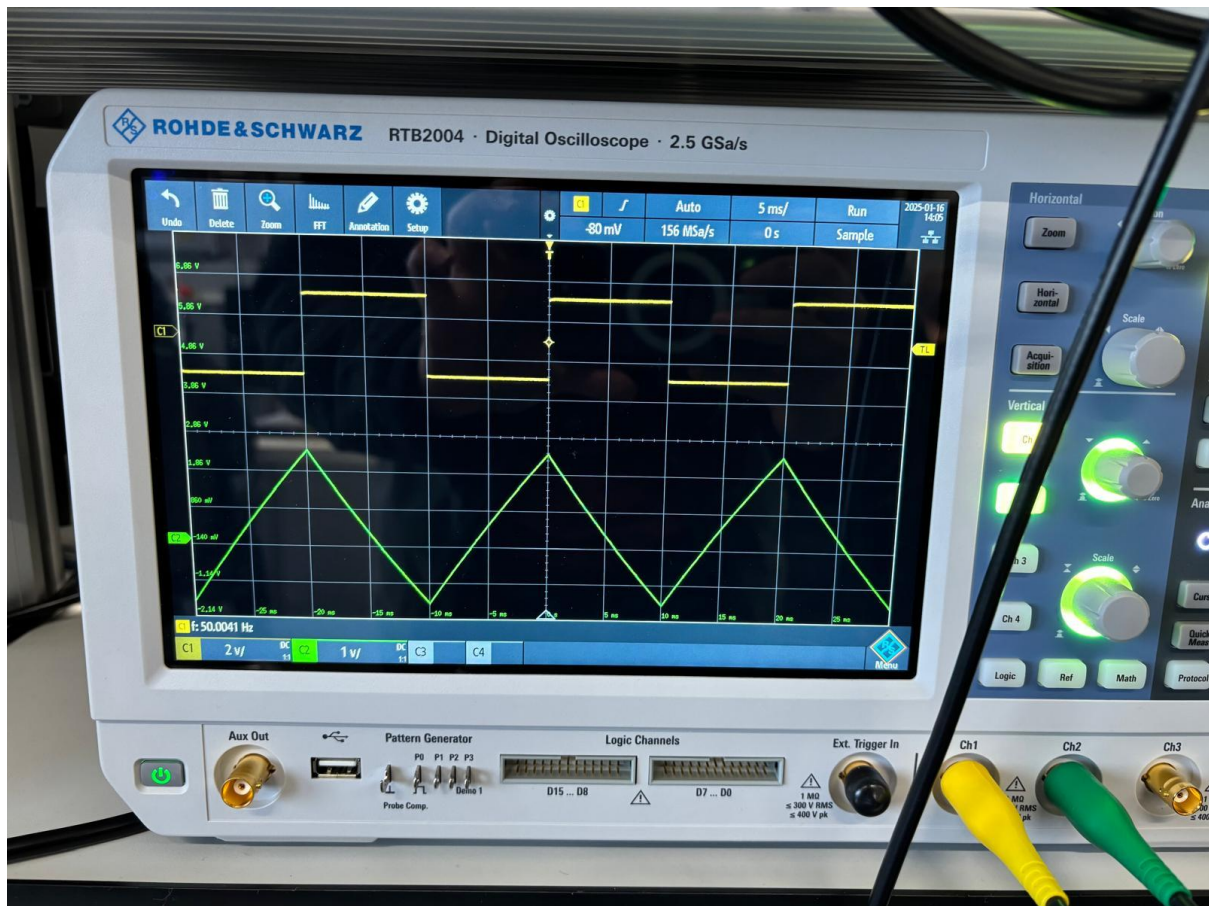
Task 2: Triangular Wave Generation

Observations:

- Oscilloscope Display (Image 2):
 - A clean triangular waveform is observed, confirming successful integration of the square wave input.
 - Frequency: $f \approx 42.1554 \text{ Hz}$ (same as square wave input).
 - Peak-to-Peak Voltage: $V_{\text{pp}} \approx 10 \text{ V}$.

Analysis:

- The triangular waveform was generated by integrating the square wave input using an operational amplifier.
- The observed triangular waveform has the same frequency as the square wave, validating the integration process.



Discussion

1. Square Wave Generation:

- The generated square wave frequency was consistent with theoretical calculations. Minor discrepancies could be due to component tolerances and experimental setup.

2. Triangular Wave Generation:

- The output triangular wave displayed the expected characteristics. The capacitor and resistor values were crucial in determining the ramp rate of the wave.